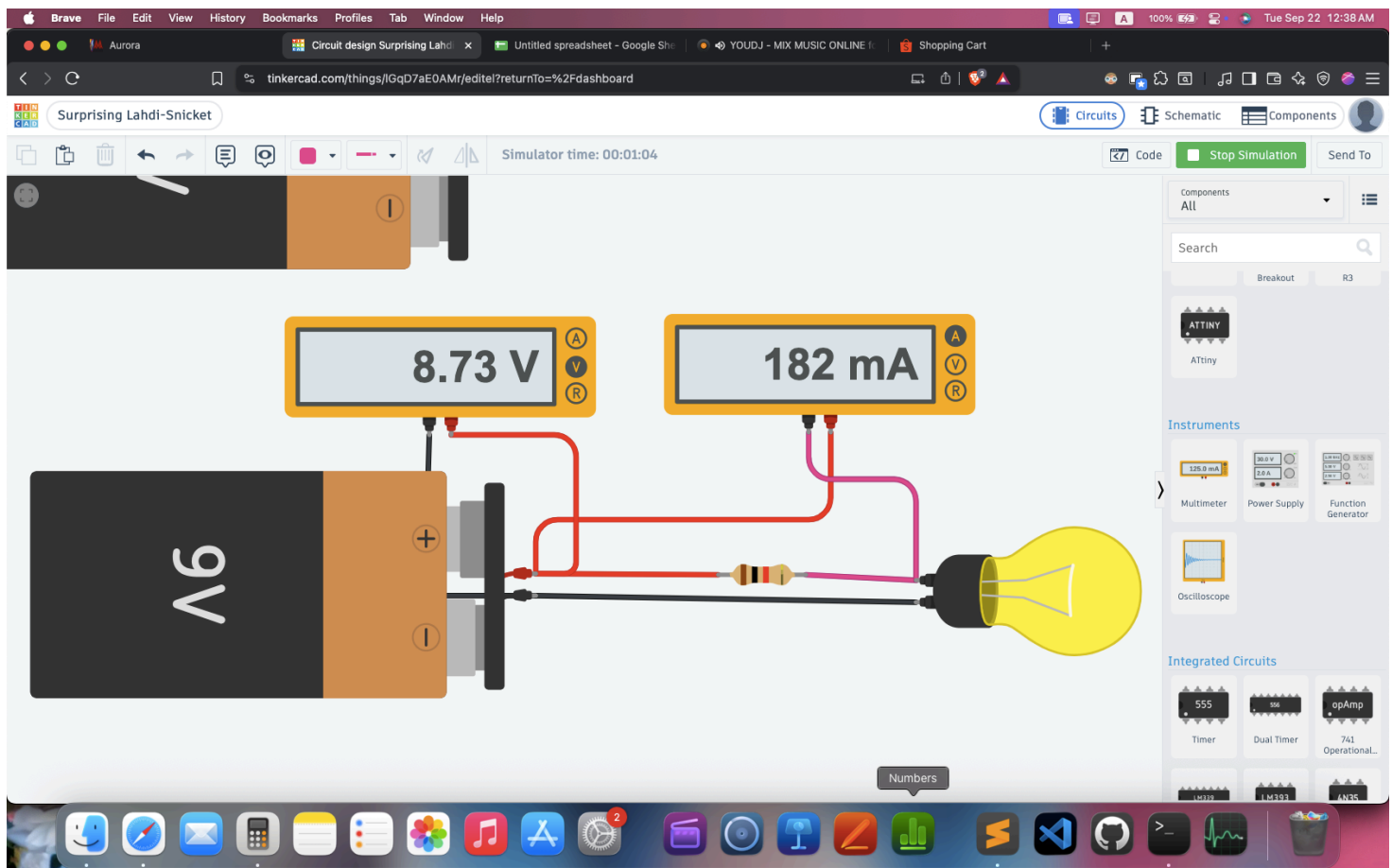


Circuit Simulated in TinkerCAD

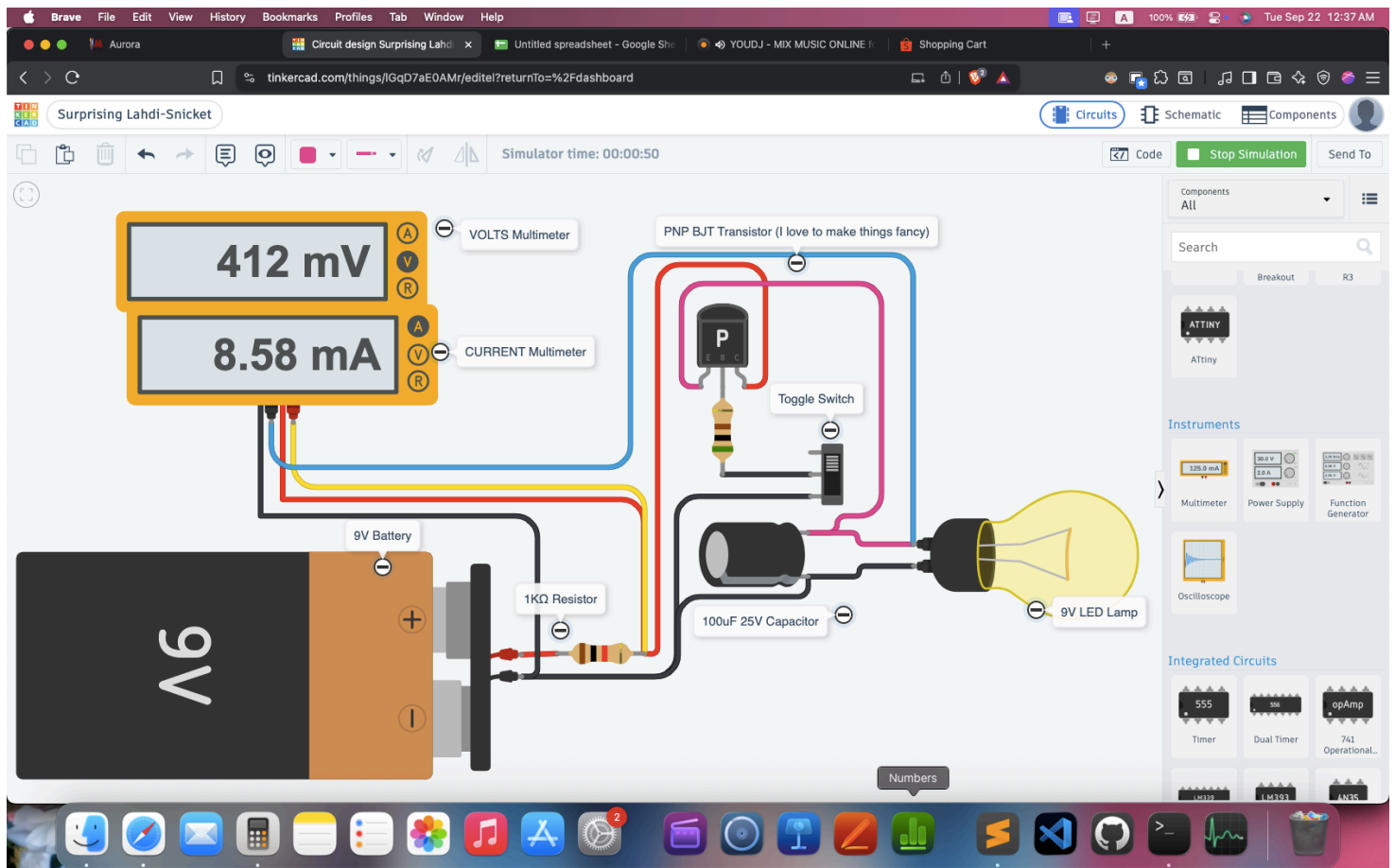
Please note that my physical prototyping boards and pre-prototyping stage components, including the breadboard and related circuit components, are currently in my Development Lab and unavailable at home. Hence, I was unable to physically build and photograph the circuit. Instead, I designed and tested the LED circuit digitally using TinkerCAD to demonstrate proper connections and operation, and prepared the circuit diagram and documentation for submission.

Basic Circuit - Conventional Design



In Rev1.0, the simple LED circuit was designed and tested digitally using Tinkercad, where the positive terminal of the 9V battery was connected to one end of the 1k Ω resistor, the other end of the resistor to the positive leg of the 9V LED lamp, and the negative leg of the LED lamp back to the negative terminal of the battery to form a complete and functional circuit. During simulation, the multimeters recorded a voltage of 8.73 V and a current of 182 mA or 0.182 A, with the LED lamp ON, indicating proper operation. Using the formula Power (P) = Voltage (V) x Current (I), the power consumed by the LED lamp is calculated as $P = 8.73 \text{ V} \times 0.182 \text{ A}$, which is approximately 1.59 watts. The underlying principle of this circuit is the conversion of chemical energy stored in the battery into electrical energy, where the resistor acts as a current limiter based on Ohm's Law to prevent damage to the LED by converting excess energy into heat, while the LED itself converts electrical energy into light energy through electroluminescence. To improve energy efficiency and minimize environmental impact, strategies include using a resistor value precisely calculated for the LED's forward current to reduce heat waste, adding a toggle switch for on and off control to prevent unnecessary energy consumption, using a high-efficiency LED lamp, using a rechargeable 9V battery, and implementing additional filtering such as a capacitor for voltage stability.

Energy-Efficient Version (My Fancy Version)



Revision 1.0 recorded 8.73 V and 182 mA with a continuous power consumption of approximately 1.59 watts and no way to turn it off without disconnecting the battery, Revision 2.0 introduces a PNP BJT transistor as a high-side switch, a toggle switch for user-controlled on and off operation, a 1kΩ resistor for precise current limiting, and a 100μF 25V capacitor for voltage stabilization, as shown in the simulation where the VOLTS multimeter reads 412 mV and the CURRENT multimeter reads 8.58 mA in the low-power control branch that drives the transistor. This design is more effective because when the toggle is off, the circuit draws zero current, eliminating standby waste and drastically extending 9V battery life from hours to days, and when on, the transistor limits the total current to the LED to a much lower and more stable level than the 182 mA direct draw in Revision 1.0 while the capacitor smooths voltage to prevent flicker and reduce stress on the battery. The result is a circuit that is more energy-efficient, more sustainable, and more practical for real-world use, as it reduces heat waste, minimizes environmental impact from frequent battery replacement, and provides reliable and controllable light output, making it a superior solution worldwide compared to a basic always-on resistor-LED circuit.